Journal of
Theoretical and Applied Electronic Commerce Research

MDPI

Article

Disconfirmation Dynamics in Service Recovery: Insights from Online Customer Reviews

Woojin Lee [1] and Junsung Park [2,*]

[1] Department of Computer Science and Artificial Intelligence, Dongguk University, Seoul 04620, Republic of Korea; wj926@dgu.ac.kr
[2] Department of Data Science, Business School, Incheon National University, Incheon 22012, Republic of Korea
* Correspondence: js.park@inu.ac.kr

Abstract

This paper analyzes the effects of service recovery and the double-deviation phenomenon on customer satisfaction using an analysis of online reviews. With the application of Word2Vec and AFINN sentiment analysis to 4793 Skytrax review comments, this paper bridges the gap between conceptual theories about customer behavior and big data analysis. Findings from this research indicate that although overall types of service recovery increase customer satisfaction, explanation is the most preferable yet effective type of recovery. Most importantly, this analysis shows that service recovery disconfirmation has been found to be a highly important moderator. When service efforts involve intangible actions, such as an apology or explanation, that fail to satisfy customers, they tend to perform worse than no recovery at all, thus supporting the double-deviation phenomenon. Compensation, however, has been found to be an efficient type of recovery since its impacts are not reduced much due to service recovery disconfirmation. These insights provide service providers with critical guidance on prioritizing transparent communication and avoiding poorly executed intangible recoveries.

Keywords: double deviation; sentiment analysis; customer satisfaction; online reviews; service recovery

1. Introduction

In the service industry, service failures are inevitable occurrences that can have a profound impact on customer satisfaction and loyalty [1]. When service providers fail to meet customer expectations, it often leads to negative experiences, resulting in customer dissatisfaction and a reduced likelihood of them returning [2]. Therefore, effective service recovery efforts are crucial in minimizing the adverse effects of service failures and in restoring customer willingness to come back.

Up to this point, research in the field of Service Failure (SFR) has predominantly focused on Behavioral SFR. Behavioral SFR heavily relies on laboratory experiments and surveys, with a particular emphasis on subjective outcomes such as attitudes and perceptions. These studies have built upon the foundation of justice theory and have explored various aspects related to word-of-mouth, customer satisfaction, forgiveness, loyalty, and the intention to revisit. For example, researchers have delved into topics like the effectiveness of different service recovery approaches [3–6], how customers can forgive service failures [7–9], whether customers will return for the service [10], or if they might become angry and engage in negative behavior towards the company due to service failure [11–16].

Check for updates

Received: 23 December 2025
Revised: 16 January 2026
Accepted: 21 January 2026
Published: 2 February 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

However, despite these comprehensive studies, a critical area called 'double deviation' still remains relatively underexplored [17,18]. According to Bitner et al. [19] and Grégoire and Fisher [12], define double deviation as the phenomenon where an inadequate recovery attempt acts as a second service failure, exacerbating customer dissatisfaction. While successful recovery efforts that meet or exceed customers' initial expectations can lead to enhanced satisfaction, recovery efforts that fall short of these expectations, it may result in negative disconfirmation. This negative disconfirmation potentially worsens the effects of the service failure, creating a double-deviation scenario.

The consequences of double deviation are severe and far-reaching in real-world contexts. A prime example is the 2017 United Airlines passenger removal incident. While the initial service failure stemmed from the forcible removal of a passenger due to overbooking, it was the company's inadequate recovery response that triggered global outrage. This failed recovery resulted in negative word-of-mouth across social media, a $1.4 billion drop in market value within days, and lasting reputational damage [20,21].

Despite its critical impact on business survival, the double-deviation phenomenon has received relatively limited attention in empirical research [17,18]. The primary reason lies in the methodological limitations of traditional approaches. Since behavioral SFR primarily relies on surveys, it faces inherent challenges in capturing the true nature of double deviation. Fundamentally, retrospective surveys suffer from recall bias, as respondents rely on fading memories of past events [22]. More importantly, controlled experiments often fail to replicate the high arousal negative emotions such as intense customer rage and feelings of betrayal that characterize real world double-deviation incidents [23]. Respondents who answer questions in a calm and controlled setting may not accurately anticipate their immediate emotional reactions to a severe service failure, resulting in a discrepancy between theoretical predictions and real-world behaviors.

To overcome the methodological constraints of survey-based research, Quantitative SFR utilizing online customer reviews has emerged as a critical alternative. Unlike self-reported data obtained from controlled experiments, online reviews provide observational data reflecting actual customer experiences in naturalistic settings [24]. With the rapid development of digital platforms, collecting large-scale datasets on service failures has become feasible, offering a more accurate reflection of customer perceptions.

However, despite these advantages, current quantitative studies often lack theoretical depth [25]. Much of the existing literature focuses on descriptive sentiment analysis or pattern recognition without incorporating established theoretical frameworks from the behavioral SFR field. While these studies effectively identify the presence of negative sentiments, they often fail to explain the underlying mechanisms that drive these reactions, such as double deviation or justice theory. Consequently, there is a critical need for research that integrates behavioral theories with big data analytics to provide actionable insights.

To bridge this gap, this study integrates hypothesis testing with big data analytics. We utilize a large-scale dataset from Skytrax to examine the impact of recovery efforts on customer satisfaction. By employing Natural Language Processing (NLP) techniques to operationalize recovery variables (apology, compensation, explanation), this research empirically validates behavioral hypotheses within a real-world context. This methodological integration offers a robust examination of how recovery efforts and disconfirmation interact to influence customer satisfaction. The specific research questions guiding this study are as follows:

(1) How do recovery efforts (apologies, compensation, explanation) influence customers' satisfaction?
(2) How does the presence of disconfirmation affect the relationship between recovery efforts and customer satisfaction?

https://doi.org/10.3390/jtaer21020045

2. Literature Review

2.1. Customer Satisfaction

Customer satisfaction refers to the emotional response individuals experience when a product or service meets or exceeds their expectations [26]. It stands as a vital factor, drawing significant attention from researchers in service-related industries. Customer satisfaction has associations with various positive factors in the service context, including loyalty [27], revisit intention [28], word-of-mouth [29].

Conversely, when satisfaction is low due to service failure, adverse outcomes emerge. Customers experiencing service failures and subsequent dissatisfaction may engage in negative behaviors such as switching, complaining, and negative word-of-mouth [30,31]. Given the potential for these behaviors to harm a company, effectively managing customer satisfaction following service failure is of utmost importance.

Traditionally, customer satisfaction was primarily measured through survey methods. Psychological variables such as satisfaction are challenging to observe externally and difficult to quantify like financial data. Hence, researchers primarily rely on survey methods that extract numerical data from individuals' responses. However, survey methods have faced persistent limitations. First, due to cost constraints, researchers can collect relatively small amounts of data, and respondents are limited to answering pre-determined questions. Second, survey instruments may not fully capture the dynamic and intricate nature of customer experiences. Third, response data only reflects a specific point in time. And lastly, respondents may be influenced by response fatigue, social desirability biases, leading questions, or respond insincerely for survey rewards [32–35].

With the recent development of social media, the availability of online reviews has grown substantially. Researchers can now observe individual psychological and cognitive variables through online reviews. Online reviews can collect responses at various time points and accurately reflect customers' true feelings and experiences. Furthermore, reviews precisely highlight the gap between customer expectations and actual experiences, making them more effective in measuring customer satisfaction than surveys [32,33]. Measuring customer satisfaction in online reviews typically involves two approaches: using ratings and analyzing customer reviews themselves. Ratings offer the advantage of being already quantified, as they are direct satisfaction assessments by customers [36]. However, critics argue that review ratings can be biased or inaccurate, as customers may give high ratings while writing negative reviews or vice versa [37,38]. In the latter approach, which relies on customer reviews, researchers believe that it is essential to understand customers' perceptions through their actual reviews. Scholars in this category apply NLP techniques to perform sentiment analysis on customer reviews, thereby measuring customer satisfaction [32,34,39]. Sentiment Analysis (SA), a subset of Natural Language Processing (NLP), allows for the evaluation of collective attitudes toward specific topics through text data [40]. Based on these discussions, this study aims to utilize sentiment analysis as a tool for measuring customer satisfaction

2.2. Service Recovery and Justice Theory

Service recovery is defined as the specific actions and efforts companies undertake to fix a problem or address a failure in meeting customer expectations regarding their services [41,42]. Preventing service failure is the best course of action, but even if prevention fails, providing appropriate recovery can manage customer dissatisfaction [43]. According to justice theory, customers evaluate distributive justice, procedural justice, and interactional justice during the recovery process [44,45]. Distributive justice refers to the compensation received by customers, such as refunds, discounts, and free gifts. Procedural justice refers to the speed of response, procedural, policies, and methods that a firm uses

to address customer complaints [6]. Interactive justice refers to the way employees treat customers, such as apologies, explanations, and empathy [12,44,46]. In line with justice theory, previous research has commonly regarded recovery strategies as involving apologies and compensation [3,6,47–50]. While apology is a verbal or written expression of regret for the service failure [3], compensation involves providing customers with financial or non-financial rewards to make up for the inconvenience caused [51].

Apology has been found to be a superior strategy for service recovery due to its empathetic nature. Empathy is a critical factor in service recovery because it allows customers to feel understood and appreciated [50]. Numerous studies have demonstrated that customers are more likely to forgive service failures when they feel that their concerns are being taken seriously and that the service provider genuinely regrets the inconvenience caused [8,47,50,52]. The ability to convey empathy through an apology can be a powerful tool in service recovery, as it can help to repair the relationship between the customer and the service provider.

In contrast, compensation, which includes monetary forms such as discounts, cashback, and refunds, may not always be sufficient to increase positive emotions like happiness and joy. However, it remains a vital means to alleviate negative emotions such as anger, disappointment, and frustration [53]. While apologies through emotional empathy can repair the damage, compensation can directly restore the reward-investment ratio through economic benefits [54,55].

In addition to the two primary recovery types, explanation has also gained recognition as an essential factor [56]. Explanations can be categorized into referential and justification [57]. Referential explanations provide detailed information about why a service failure occurred, with a focus on helping customers understand the situation through information provision. On the other hand, justification explanations aim to provide reasons that legitimize why the service failure occurred. Airline service providers employ both types of recovery strategies. For example, an explanation like "The flight was canceled due to weather conditions" falls under the referential category. By providing this detailed explanation, the airline aims to help passengers understand that the cancelation was necessary to ensure their safety and the safety of the flight crew. In contrast, an explanation like "The flight is delayed due to mechanical maintenance" falls under the justification category. It provides reasons to support the airline's decision to delay the flight, reassuring passengers that their safety is the top priority and that taking the time to address the technical problems is in their best interest.

2.3. Disconfirmation and Customer Behavior

According to Expectancy Disconfirmation Theory (EDT), disconfirmation refers to the gap between people's expectations and their actual experiences. If the actual experience is significantly better than expected, individuals tend to develop a favorable attitude towards the product or service. Conversely, if a negative gap exists, individuals become dissatisfied and may reconsider their relationship with the service provider [58].

While SFR researchers have focused on what recovery strategies companies should employ when service failures occur, there has been limited attention on customer behavior after the recovery process. Some researchers have highlighted that ineffective actions or remedies by service providers, such as the failure to acknowledge, apologize, or compensate the customer adequately, can lead to double [19,59]. Furthermore, these flawed recovery measures impact customer perceptions and appraisals, including evaluations of distributive, procedural, and interactional justice, which have emerged as significant factors influencing double deviation [60,61]. When customers perceive that their sense of justice has been violated due to the service provider's ineffective recovery actions, it

can nullify the effectiveness of the recovery effort itself. Considering that the primary aim of recovery is to restore trust and satisfaction with customers following a service failure, recovery disconfirmation can undermine these efforts. Consequently, in the presence of disconfirmation, the recovery efforts provided by the company may fail to fully restore customer satisfaction.

3. Hypothesis Development

3.1. Effects of Recovery Efforts on Satisfaction

Recovery efforts are not merely administrative procedures but are strategic interventions that reshape customer perceptions following a failure. Empirical research grounded in Justice Theory consistently demonstrates that specific recovery attributes, directly influence post-failure satisfaction by restoring perceived fairness.

An apology serves as a critical mechanism for restoring interactional justice [44]. Unlike tangible rewards, an apology addresses the emotional deficit caused by a service failure. Roschk and Kaiser [50] empirically demonstrated that apologies conveying authentic empathy significantly reduce customer aggression and enhance satisfaction. Similarly, Abeler et al. [3] found that even costless apologies could yield positive economic outcomes by restoring trust. Radu et al. [5] further supported this by showing that apologies, when perceived as sincere, are often more effective than material compensation in repairing relationships. Based on these empirical findings, we posit that an apology will positively influence customer satisfaction.

H1. *Apologies positively affect satisfaction.*

Compensation targets distributive justice by offering tangible value to offset the costs incurred by the customer. Drawing on Equity Theory [54], compensation serves to restore the balance between a customer's input and output. Wirtz and Mattila [6] identified that compensation is effective because it restores the equity of the exchange relationship, balancing the failure with a tangible gain. Furthermore, Roschk and Gelbrich [51] conducted a meta-analysis revealing that monetary compensation is particularly effective in alleviating immediate dissatisfaction and negative emotions such as frustration. Grewal et al. [49] also found that appropriate compensation significantly increases repurchase intentions by mitigating the perceived loss. Thus, we expect that providing compensation will have a direct positive effect on satisfaction.

H2. *Compensation positively affects satisfaction.*

Explanation functions as a vital tool for informational justice, a key dimension of interactional justice. Wang and Mattila [56] argued that providing a clear reason for a failure reduces customer uncertainty and perceived loss of control. Drawing on Attribution Theory, Sparks and Fredline [57] empirically showed that when companies provide a referential explanation or a justification, customers are less likely to attribute the failure to the firm's negligence, thereby reducing anger. By offering transparency, explanations help customers rationalize the incident, which in turn fosters higher satisfaction compared to unexplained failures.

H3. *Explanation positively affects satisfaction.*

3.2. The Moderating Role of Disconfirmation

While recovery efforts are generally beneficial, their efficacy is contingent upon the disconfirmation of expectations. Grounded in the framework proposed by Bitner et al. [19],

prior literature suggests that a recovery attempt that fails to meet customer expectations operates as a distinct, secondary failure, potentially nullifying the positive effects of the recovery itself.

Grégoire and Fisher [11,12] established that when high-relationship customers perceive a recovery effort as betrayal or insufficient, their desire for retaliation increases significantly. If an apology is offered but perceived as robotic, insincere, or delayed, it violates interactional justice. Lee et al. [13] further confirmed that such double deviations can exacerbate negative emotions, implying that a flawed apology might be worse than no apology at all. Therefore, we argue that disconfirmation will negatively moderate the positive link between apology and satisfaction.

H4. *Recovery disconfirmation negatively moderates the relationship between apology and customer satisfaction.*

Although compensation is a powerful recovery tool, its impact is heavily contingent upon customer expectations regarding fairness. Drawing on Equity Theory [54], customers evaluate compensation by comparing their input with the output. Wirtz and Mattila [6] highlight that compensation is effective only when it restores the 'equity' of the exchange. However, Smith et al. [44] established that when the compensation is perceived as inadequate compared to the loss, it triggers a sense of distributive injustice. In such cases, the intended positive effect of compensation is diminished because the customer focuses on the inequity of the offer rather than the economic value received. Thus, we hypothesize that disconfirmation will dampen the positive effect of compensation.

H5. *Recovery disconfirmation negatively moderates the relationship between compensation and customer satisfaction.*

For explanations, the gap between expectation and reality is critical. Bies and Shapiro [62] established that for an explanation to be effective, it must be perceived as adequate and sincere. Otherwise, it is dismissed as a just excuse. If an explanation is expected but not provided, or if the provided explanation is vague and defensive, it fails to reduce uncertainty and violates informational justice.

Bitner et al. [19] highlighted that such negative employee responses are primary triggers of the double-deviation effect. In these instances, the explanation attempts are perceived as an evasion of responsibility, which exacerbates the initial dissatisfaction rather than resolving it. Consequently, we propose that the positive impact of explanation on satisfaction will be weakened when disconfirmation is present.

H6. *Recovery disconfirmation negatively moderates the relationship between explanation and customer satisfaction.*

4. Research Methodology

To empirically investigate the relationship between service recovery efforts, expectancy disconfirmation, and customer satisfaction, this study proposes a comprehensive framework utilizing unstructured online review data. Unlike traditional approaches utilizing survey data, this research adopts a data-driven approach by transforming unstructured text into structured variables. Specifically, we employ natural language processing to identify and quantify key constructs such as recovery efforts and disconfirmation directly from customer reviews. Following the extraction process, these variables serve as inputs for the regression model to test the proposed hypotheses. The overall analytical framework is illustrated in Figure 1.

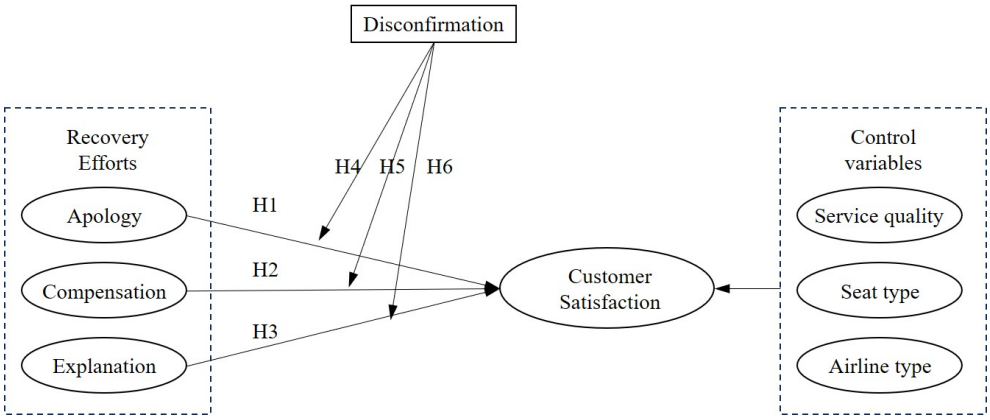


Figure 1. Model framework.

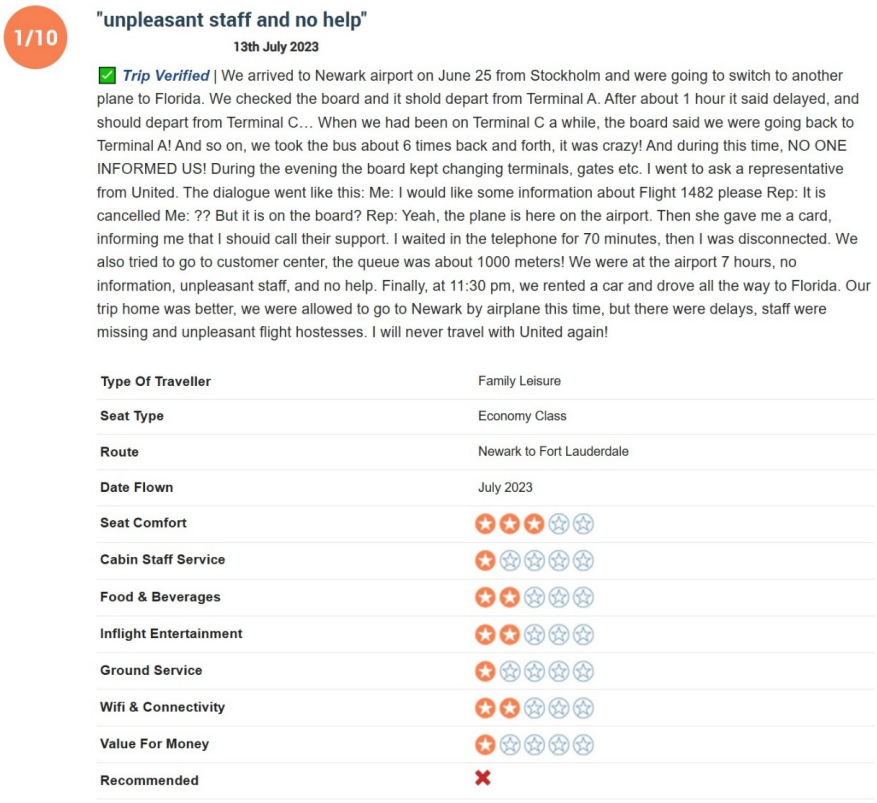
As illustrated in Figure 1, the proposed model framework visualizes the causal pathways between service recovery strategies and customer satisfaction. The model posits three direct paths where specific recovery actions are hypothesized to positively influence satisfaction (H1, H2, H3). Moreover, the framework incorporates disconfirmation as a moderator to capture the dynamic nature of service recovery. The interaction paths (H4, H5, H6) examine whether the effectiveness of these recovery efforts is diminished when the recovery fails to meet customer expectations, empirically testing the double-deviation phenomenon. Additionally, control variables are integrated into the model to account for exogenous factors influencing customer satisfaction.

4.1. Data Description

We collected review data from Skytrax (http://www.airlinequality.com/ (accessed on 17 July 2023)) during the period of 17–21 July 2023. Data collection was performed using an automated web scraping script written in Python 3.12, designed to retrieve all available historical reviews without manual intervention. Unlike platforms that deal with travel in general, such as TripAdvisor or Expedia, Skytrax exclusively focuses on the airline industry, allowing us to access a large volume of data [63]. While historical records exist, we restricted our dataset to the period from May 2015 to July 2023. This restriction aligns with a major platform update that standardized the review submission interface. Focusing on reviews generated under this unified framework ensures the structural homogeneity of both the unstructured text and the quantitative ratings. Skytrax not only provides review content but also offers additional information, including the type of aircraft flown, seat type, date of travel, and service quality variables such as Food and Beverages, Inflight Entertainment, Seat Comfort, Staff Service, Value for Money, Cabin Staff Service, Ground Service, and Wifi and Connectivity. These variables were collected to serve as control variables in our analytical model, accounting for factors influencing satisfaction beyond the recovery process. An example of an online review is illustrated in Figure 2.

Given our research's focus on examining relationship between recovery efforts and satisfaction, we collected review titles, review content, ratings, and additional airline type which whether it is LCC or FCC, the eight service quality variables, and seat type data for control variables. To ensure the credibility of the reviews, we collected reviews from all airlines with over 500 reviews posted from May 2015 to July 2023. This resulted in a total of 27 airlines and 72,860 reviews. As our study aims to understand how customers perceive overall service satisfaction when a company's recovery response falls short of expectations following a service failure, we chose to extract only reviews where service failures were experienced. Since the most critical core service failures in the airline industry

are cancelations and delays [64], we filtered reviews specifically mentioning cancel and delay.


1/10

"unpleasant staff and no help"

13th July 2023

Trip Verified | We arrived to Newark airport on June 25 from Stockholm and were going to switch to another plane to Florida. We checked the board and it shold depart from Terminal A. After about 1 hour it said delayed, and should depart from Terminal C… When we had been on Terminal C a while, the board said we were going back to Terminal A! And so on, we took the bus about 6 times back and forth, it was crazy! And during this time, NO ONE INFORMED US! During the evening the board kept changing terminals, gates etc. I went to ask a representative from United. The dialogue went like this: Me: I would like some information about Flight 1482 please Rep: It is cancelled Me: ?? But it is on the board? Rep: Yeah, the plane is here on the airport. Then she gave me a card, informing me that I shouid call their support. I waited in the telephone for 70 minutes, then I was disconnected. We also tried to go to customer center, the queue was about 1000 meters! We were at the airport 7 hours, no information, unpleasant staff, and no help. Finally, at 11:30 pm, we rented a car and drove all the way to Florida. Our trip home was better, we were allowed to go to Newark by airplane this time, but there were delays, staff were missing and unpleasant flight hostesses. I will never travel with United again!

Type Of Traveller	Family Leisure
Seat Type	Economy Class
Route	Newark to Fort Lauderdale
Date Flown	July 2023
Seat Comfort	★★★☆☆
Cabin Staff Service	★☆☆☆☆
Food & Beverages	★★☆☆☆
Inflight Entertainment	★★☆☆☆
Ground Service	★☆☆☆☆
Wifi & Connectivity	★★☆☆☆
Value For Money	★☆☆☆☆
Recommended	✖

Figure 2. Review example (Stars represent individual service ratings on a five-point scale, and the 'X' symbol indicates that the flight is not recommended to others).

To achieve this, we employed the Word2Vec method. The Word2Vec model learns word embeddings, representing words as dense vectors in a high-dimensional space [65]. These vector representations capture semantic and syntactic relationships between words. Cosine similarity between word vectors determines the similarity between words. Words with similar vectors, indicating similar contextual usage, have higher cosine similarity scores and are considered more similar.

To ensure the quality and consistency of the collected data, a series of preprocessing steps were applied to the dataset. These steps involved handling missing values, expanding contractions, removal of stop words, lemmatization and tokenization. Since Word2Vec operates on word relationships, all sentences needed to be tokenized. To standardize textual data and reduce dimensionality, lemmatization was applied to the words in the reviews.

Furthermore, we utilized the n-gram method to identify similar expressions. N-grams represent continuous sequences of n words forming a single semantic unit. After consolidating duplicate or similar meanings, we identified the following similar words: 'cancel', 'delay', 'reschedule', 'delayed', 'change_flight', 'miss_connect_flight', 'flight_change', 'rebook', 'rebooking', 'flight_late', 'cancellation', 'get_delayed', 'rebooked', 'get_cancel', 'affected_delay', 'change_time', 'plane_delayed', 'late_flight', 'postpone'. Reviews containing these words were extracted, resulting in 23,893 reviews. Next, we examined the recovery efforts received by individuals who had experienced service failures. Due to the nuanced nature of language, the mere presence of specific words does not necessarily imply the receipt of the corresponding recovery. Therefore, we engaged two doctoral-level researchers and two master's-level researchers to collectively label the specific recoveries received. Finally, we extracted 4793 reviews for our analysis.

4.2. Measurements

In this study, we utilized the AFINN lexicon score to measure customer satisfaction. AFINN is a lexicon of words manually rated for valence, ranging from minus five (negative) to five (positive), created by Finn Årup Nielsen between 2009 and 2011 [66]. AFINN assigns a predefined sentiment score to each word, and the total AFINN score for a sentence is calculated as the sum of the sentiment scores of the words in the sentence. Consequently, the total score varies based on sentence length and the emotional intensity of the words. Given that other variables in this study have relatively restricted ranges, we normalized the customer satisfaction values through z-score normalization to bring them within a range of 1 (positive) to −1 (negative) to mitigate value disparities.

For recovery efforts, we measured three types, apology, compensation, and explanation. These are binary variables with a value of '1' indicating that the specific recovery effort was received and '0' indicating that there was no recovery effort. Specifically, if apology was conveyed by cabin crew or through a notification message, it was coded as '1', while if there was no apology, it was coded as '0'. Regarding compensation, in case of delay or cancelation the provision of additional financial compensation, such as meal coupons or hotel accommodation, was coded as '1'. If no compensation was offered, this was coded as '0'. For explanation, the criterion was whether the airline provided a reason for the delay or cancellation. If a reason was provided, it was coded as '1'. If the airline merely informed passengers of the delay without an explanation, or if passengers had to ascertain the reason themselves, it was coded as '0'.

Prior studies on disconfirmation primarily utilized survey methods to measure it. For example, some researchers indirectly inferred service failure during the recovery process by measuring customer recovery satisfaction [16,19], set specific disconfirmation scenarios [13], or directly assessed the gap between customer expectations and actual recovery through surveys [67]. However, since disconfirmation represents the gap between customer expectations and actual service, survey experiments may introduce errors due to customers potentially misinterpreting and comparing abstract concepts accurately.

In contrast, through online reviews, we can directly examine how individuals who received a specific recovery evaluated it, along with the type of service failure they experienced. Hence, we considered disconfirmation to be present when specific reviews indicated that they experience recovery but expressed dissatisfaction with it. For example, in the case of an apology, reviews like "They apologized, but it was insincere and rude" were deemed to exhibit disconfirmation. Regarding compensation, there were two disconfirmation types: receiving low-value financial compensation or being promised compensation in case of delay or cancelation but not receiving it. For explanation, disconfirmation was identified when the airline failed to provide a proactive explanation of the reason for delay or cancellation, and passengers had to inquire or were left without information. In summary, these disconfirmations were coded as 1 if present and 0 if absent, resulting in binary variables.

To account for factors that prior research has identified as influencing customer satisfaction, we included several control variables. These encompassed the airline type that passengers flew with and the type of seats they purchased. People tend to form expectations based on the price they paid, influencing their level of satisfaction [33,34]. Therefore, we added airline type (consisting of 27 different airline types) and seat type (Economy, Premium Economy, Business, First Class) as control variables. Moreover, the primary goal of this study is to track perceptual changes among individuals during the recovery process following a service failure. Consequently, factors unrelated to service failure and recovery, specifically service quality factors such as Food and Beverages, Inflight Entertainment, and Wi-Fi and Connectivity, are utilized as control variables. Service quality was measured by averaging the values of the three collected variables.

4.3. Model

To test our hypotheses, we employed a multiple regression model with customer satisfaction as the dependent variable, service quality, three recovery efforts, and disconfirmation as independent variables, and airline type and seat type as control variables. The model is structured as follows:

$$CS_{ij} = b_i + AP_{ij} + CP_{ij} + EX_{ij} + DC_{ij} + AP_{ij} \times DC_{ij} + CP_{ij} \times DC_{ij} + EX_{ij} \times DC_{ij} + SQ_{ij} + AT_{ij} + ST_{ij} + e_{ij}$$

where CS_{ij} = customer satisfaction of review of airline j, AP_{ij} = whether an apology was received for the service failure of review i of airline j, CP_{ij} = whether a compensation was received for the service failure of review i of airline j, EX_{ij} = whether an explanation was received for the service failure of review i of airline j, DC_{ij} = whether a disconfirmation occurred during review i of airline j, SQ_{ij} = service quality of review i of airline j, AT_{ij} = airline type of review i of airline j, ST_{ij} = seat type of review i of airline j.

5. Results

5.1. Descriptive Statistics

Table 1 presents the descriptive statistics for the key variables in our analysis. The final sample consisted of 4793 reviews from customers who experienced service failures (flight delays or cancelations) across 27 airlines. Customer satisfaction, measured through AFINN sentiment analysis and normalized to a range of −1 to 1, showed a mean of −0.036 (SD = 0.177), indicating slightly negative sentiments overall among customers who experienced service failures. Regarding recovery efforts, only 4.9% of reviews mentioned receiving an apology (n = 235), while 23.9% reported receiving compensation (n = 1146), and 14.9% indicated that an explanation was provided (n = 714). Compensation was the most frequently observed recovery effort, reflecting airlines' tendency to prioritize tangible remedies such as meal vouchers and hotel accommodations over verbal acknowledgments. Disconfirmation was present in 33.9% of the sample (n = 1625). This substantial proportion underscores the prevalence of the double-deviation phenomenon in real-world service recovery contexts, indicating that roughly one-third of customers who experienced service failures also perceived the subsequent recovery attempts as falling short of their expectations. Service quality ratings, normalized to a 0–1 scale, averaged 0.272 (SD = 0.296), and the majority of respondents traveled in economy class (mean seat type = 0.203).

Table 1. Descriptive statistics of the key variables.

Variables	Mean	SD	Min	Max
Customer Satisfaction	−0.036	0.177	−1.00	1.00
Apology (AP)	0.049	0.215	0	1
Compensation (CP)	0.239	0.426	0	1
Explanation (EX)	0.149	0.356	0	1
Disconfirmation (DC)	0.339	0.473	0	1
Service quality	0.272	0.296	0	1
Seat Type	0.203	0.613	0	3

5.2. Hypothesis Testing

To test our hypotheses, we employed regression analysis with customer satisfaction as the dependent variable. Table 2 presents the complete regression results, including direct effects of recovery efforts (H1–H3) and the moderating effects of disconfirmation (H4–H6).

Table 2. Results of hypotheses test.

Variables	Coefficient	Std.err	t-Value	*p*-Value
Direct effects				
Apology (AP)	0.174	0.016	10.916	0.000 ***
Compensation (CP)	0.132	0.013	10.396	0.000 ***
Explanation (EX)	0.198	0.014	14.550	0.000 ***
Disconfirmation (DC)	−0.063	0.084	−0.747	0.455
Moderating effects				
AP × DC	−0.189	0.087	−2.177	0.030 **
CP × DC	−0.068	0.085	−0.796	0.426
EX × DC	−0.177	0.085	−2.077	0.038 *
Control variables				
Service quality	0.011	0.002	5.276	0.000 ***
Airline type (AT)	0.001	0.000	2.606	0.009 **
Seat type (ST)	0.001	0.005	0.191	0.848
Intercept	−0.085	0.010	−8.563	0.000 ***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

H1 posited that apology has a positive effect on customer satisfaction. The results support this hypothesis, revealing a significant positive relationship between apology and customer satisfaction ($\beta = 0.174,\ p = 0.000 ***$). This finding indicates that when airlines provide apologies following service failures, customers report higher satisfaction levels compared to when no apology is offered.

H2 proposed that compensation has a positive effect on customer satisfaction. This hypothesis is also supported ($\beta = 0.132\ ,\ p = 0.000 ***$). Customers who received compensation—such as meal vouchers, hotel accommodations, or travel credits—expressed significantly higher satisfaction than those who did not receive any form of compensation.

H3 suggested that explanation has a positive effect on customer satisfaction. The results strongly support this hypothesis ($\beta = 0.198,\ p = 0.000 ***$). Notably, explanation exhibited the largest coefficient among the three recovery efforts, suggesting that providing clear reasons for service failures (e.g., weather conditions, mechanical issues) is particularly effective in maintaining customer satisfaction.

Comparing the magnitude of the coefficients, the effectiveness of recovery efforts can be ranked as follows: explanation ($\beta = 0.198$) > apology ($\beta = 0.174$) > compensation ($\beta = 0.132$). This hierarchy suggests that intangible recovery efforts focused on communication and acknowledgment may be more impactful than tangible financial remedies in the airline service context.

H4 predicted that recovery disconfirmation moderates the relationship between apology and customer satisfaction. The results support this hypothesis, showing a significant negative interaction effect ($\beta = -0.188,\ p = 0.029 *$). This finding indicates that when customers perceive an apology as insincere, insufficient, or poorly delivered, the positive effect of the apology is substantially diminished. The combined effect ($0.174 - 0.189 = -0.015$) suggests that a disconfirmed apology may result in lower satisfaction than providing no apology at all, providing empirical support for the double-deviation phenomenon.

H5 proposed that recovery disconfirmation moderates the relationship between compensation and customer satisfaction. Contrary to our expectation, this hypothesis is not supported ($\beta = -0.07,\ p = 0.426$). The non-significant interaction effect suggests that compensation maintains its positive impact on customer satisfaction regardless of whether customers perceive it as meeting their expectations. This finding indicates that compensation may be a more "robust" recovery strategy, as its effectiveness is less susceptible to the double-deviation effect.

H6 predicted that recovery disconfirmation moderates the relationship between explanation and customer satisfaction. The results support this hypothesis ($\beta = -0.177$, $p = 0.038*$). When customers perceive explanations as inadequate, unclear, or provided too late, the positive effect of explanation is significantly weakened. Similarly to apology, the combined effect (0.198 – 0.177 = 0.021) indicates that a poorly executed explanation barely improves satisfaction compared to no explanation at all.

5.3. Robustness Checks

To ensure robustness, we compared three models (Model 1 to Model 3). Model 1 in-cluded only customer satisfaction and control variables. In Model 2, we added recovery efforts as a main effect to Model 1. Model 3, our proposed model, introduced interaction effects between disconfirmation and recovery efforts. The log-likelihood and AIC calculations revealed that the proposed model is the most suitable among the three models. De-tailed results are presented in Table 3.

Table 3. Results of robustness checks.

Model	Description	Log-Likelihood	AIC
Model 1	Customer satisfaction and control variables	1531.774	−3053.548
Model 2	Model 1 + Recovery efforts	1765.243	−3506.486
Model 3	Model 2 + Disconfirmation (Proposed model)	1531.774	−3053.548

6. Discussion

This study explored the dynamics of disconfirmation during the service recovery process, using online customer reviews. The results can be summarized as follows. First, this study revealed that all three recovery efforts have a significant positive effect on customer satisfaction. These results underscore the importance of providing appropriate recovery actions when service failures occur, consistent with previous research [7,50,53]. However, the magnitude of effects varied considerably across recovery types, warranting closer examination.

Among these strategies, explanation emerged as the most effective recovery strategy (β = 0.198), outperforming both apology (β = 0.174) and compensation (β = 0.132). This finding aligns with Weiner's [68] attribution theory, which posits when customers receive clear explanations for service failures, they can attribute the cause to external, situational factors. According to Folkes [65], such external attributions reduce perceived intentionality and blame, thereby mitigating negative emotional responses. Furthermore, as Sparks and Fredline [69] suggest, explanations address customers' fundamental need for cognitive closure, understanding why something happened helps restore a sense of predictability and control. In the airline context, where delays and cancelations often stem from safety-related factors, transparent explanations may signal that the airline prioritizes passenger welfare over operational convenience.

Apology showed the second strongest effect (β = 0.174), which supports interactional justice perspectives emphasizing the importance of respectful, empathetic treatment during service recovery [12,44,46]. As Roschk and Kaiser [50] demonstrate, apologies serve a symbolic function. They acknowledge wrongdoing, validate customer emotions, and signal the company's commitment to the relationship. Hu et al. [70] further argue that perceived sincerity is the key mechanism through which apologies influence recovery satisfaction.

Interestingly, compensation demonstrated the smallest direct effect (β = 0.132) among the three recovery types. This finding contrasts with economic exchange perspectives rooted in equity theory, which suggest that tangible remedies should be most effective in restoring the input-output balance [54,55]. Several explanations may account for this result. First, in the airline industry, standard compensation (meal vouchers, hotel accommodations) may be perceived as routine procedural responses rather than genuine attempts to address customer concerns. Second, as Valentini et al. [53] mentioned, compensation alone without accompanying explanation or apology, may feel transactional and impersonal. Therefore, it does not address the emotional and psychological dimensions of service failure. Third, the relatively smaller effect of compensation aligns with Roschk and Kaiser's [50] findings that psychological recovery efforts often outperform economic ones in restoring customer satisfaction. Our results extend this finding to real-world online review contexts.

The most theoretically significant contribution of this study concerns the moderating role of disconfirmation. The results indicate that recovery strategies differ in how they are affected by double deviation. This refers to a situation where unsuccessful recovery efforts lead to outcomes even worse than the first failure [12,19,59].

For apology, disconfirmation produced a negative moderating effect (β = −0.189, $p = 0.030$). The combined effect under disconfirmation conditions (0.174 − 0.189 = −0.015) indicates that a disconfirmed apology actually harms customer satisfaction relative to providing no apology. This finding can be understood through expectancy violation theory [71]. When customers receive an apology, they form expectations about the company's acknowledgment of responsibility and commitment to remediation. If an apology perceived as insincere, scripted, or dismissive violates these expectations, signaling that the company either does not genuinely care or is merely engaging in superficial damage control. Therefore, such perceived insincerity may be interpreted as a form of disrespect, triggering stronger negative reactions than simply receiving no apology.

Similarly, explanation showed significant vulnerability to disconfirmation (β = −0.177, $p = 0.038$). The combined effect (0.198 − 0.177 = 0.021) indicates that a disconfirmed explanation barely improves satisfaction. This pattern aligns with Wang and Mattila's [56] regarding informational fairness. Their research revealed that explanations are inherently informational and cognitive in nature. When an explanation is perceived as inadequate, it fails to provide the cognitive closure that customers seek. Worse, as Sparks and Fredline [57] argue, an inadequate explanation may signal that the company is withholding information or being deceptive, transforming what should be a trust-building communication into evidence of untrustworthiness. In the airline context, customers who receive vague explanations like "operational reasons" without further detail may perceive this as a lack of transparency, intensifying their frustration.

The most unexpected finding is that compensation was not significantly moderated by disconfirmation (β = −0.068, $p = 0.426$). This result diverges from our theoretical expectations. Several mechanisms may explain compensation's resilience to the double-deviation effect. First, compensation may operate through a different psychological pathway than intangible recovery efforts. While apology and explanation primarily address interactional justice concerns, compensation addresses distributive justice through economic restoration [44,45]. As Kwon and Jang [72] demonstrate, even inadequate compensation still provides some restoration of the reward-investment ratio, preventing the complete nullification of recovery benefits. Second, from an attribution perspective, inadequate compensation is more likely attributed to organizational constraints rather than personal disregard [68,69]. In contrast, a poor apology or explanation is more easily attributed to the individual employee's attitude or the company's indifference toward customers. This attribution difference may explain why compensation failures feel less personally offensive.

Taken together, our findings suggest a risk-return tradeoff in service recovery strategy selection. Intangible recovery efforts offer higher potential returns, but carry substantial risk of backfiring when poorly executed. Compensation offers more modest returns but provides a safer option with minimal downside risk. This pattern has important implications for understanding the gap between research recommendations and industry practice revealed in our descriptive statistics: airlines predominantly rely on compensation (23.9%) over apology (4.9%) and explanation (14.9%). Service providers may implicitly recognize that intangible recovery requires skilled execution and carries greater risk, leading them to default to tangible compensation as a procedurally straightforward, lower-risk response.

6.1. Theoretical Implications

First, this study provides empirical validation of the double-deviation effect using real-world data, addressing a critical gap in the service failure literature. As mentioned earlier, double deviation remains empirically underexplored despite its theoretical importance [73]. Methodological constraints drive this gap, as retrospective surveys suffer from memory distortion and scenario-based experiments cannot fully capture the intense emotional impact of actual failed recoveries [25]. By utilizing online reviews, this study overcomes the limitations of prior approaches and demonstrates that behavioral theories can be systematically examined using naturally occurring textual data. This methodological contribution opens new insights for validating abstract psychological constructs in service research.

Second, this study reveals that different recovery strategies exhibit differential vulnerability to the double-deviation effect. While previous studies have examined apology and compensation [13,50,67], and some have incorporated disconfirmation as a variable [13,67], none has systematically compared how the double-deviation effect operates across multiple recovery types. Our findings show that intangible recovery efforts are highly susceptible to disconfirmation. When recovery efforts are not appropriately delivered, they may result in outcomes that are worse than offering no recovery at all.

Third, this study advances our understanding of compensation as a recovery strategy. Prior research consistently emphasized the superiority of psychological recovery efforts, with meta-analytic evidence suggesting that sincere apologies outperform monetary compensation [50,53]. In contrast, tangible recovery maintains its positive effect regardless of whether it meets customer expectations. This differential pattern suggests a risk-return framework for understanding service recovery. Intangible strategies offer higher potential returns when executed well but carry substantial downside risk, whereas compensation provides a safer, more robust option with minimal risk of backfiring. Therefore, for theoretical development, this finding suggests that the effectiveness of recovery strategies should be evaluated not only by their maximum potential impact but also by their risk profile under varying execution quality.

6.2. Practical Implications

First, it is crucial for service providers to offer explanations as a foremost response when service failures occur. The study's findings reveal that although all recovery actions are effective, explanations significantly enhance customer satisfaction. Therefore, managers should not only instruct their staff to provide detailed and transparent explanations when service failures occur but also develop standard operating procedures that guide employees in how to effectively communicate these explanations. Furthermore, in cases where service failures are anticipated, proactively notifying customers about the situation before it occurs may be more effective [55]. However, our descriptive findings reveal a notable gap between research recommendations and industry practice. Only 14.9% of customers in our sample

reported receiving an explanation, and merely 4.9% received an apology, while 23.9% received compensation. This suggests that airlines may be over-relying on tangible remedies while underutilizing the most effective recovery strategies. Service providers should rebalance their recovery portfolios to include more proactive communication.

Second, service providers should be acutely aware of the double deviation risk when executing recovery efforts. Our study empirically confirmed that intangible recovery efforts can result in outcomes worse than providing no recovery at all. This finding presents a practical dilemma in that explanation and apology possess the greatest potential to restore customer satisfaction while simultaneously carrying the highest risk of negative effects. In contexts where meeting customer expectations is difficult or where employees lack sufficient communication training, monetary compensation represents a comparatively safer recovery option.

Finally, service providers should continuously monitor online customer reviews. The digital age has made it possible for customers to instantly share their experiences, both good and bad, offering airlines a wealth of real-time data. Our finding that 33.9% of customers experienced disconfirmation highlights the prevalence of inadequate recovery in practice. Service providers should invest in advanced analytics tools and perhaps even consider setting up dedicated customer experience teams responsible for mining these online reviews for actionable insights. This way, service providers can adapt their service recovery strategies in real-time, allowing for a more agile and responsive management approach. This capacity for quick adaptation is particularly valuable in an industry marked by volatility and rapidly changing operational conditions.

6.3. Limitations and Future Research

This study primarily focuses on validating interaction effects based on online review data. While the field of machine learning, including text mining and natural language processing, generally boasts excellent predictive performance, it is often criticized for its difficulty in interpreting results, often referred to as the "black box" problem. Previous studies in the service sector using customer reviews aimed to identify service quality factors. However, analyzing interaction effects through machine learning still presents challenges. In this study, data collection and preprocessing were conducted using natural language processing methods such as text mining and word2vec. However, during the hypothesis testing phase, a binary variable distinguishing whether recovery was received or not was used in regression models. In the future, if machine learning can be used to analyze interaction effects, we anticipate more intriguing and novel results.

For online reviews, researchers often face challenges in securing data that aligns with their desired hypotheses. For instance, in this study, due to data limitations, two types of explanations (referential and explanation) were combined for analysis. However, previous research based on surveys argued that these two types lead to different attributes of corporate responsibility and, consequently, varying effects of customer anger in response to service failures [57]. Therefore, with more detailed data collection in the future, additional analyses may become feasible. Additionally, in the real world, it is common for individuals to receive more than one type of recovery simultaneously. For example, individuals may receive both an apology and compensation or an explanation and an apology simultaneously. However, considering all these interaction effects requires a more significant amount of data. In future research, creating models that account for such interaction effects with a larger dataset can provide a deeper understanding of how service recovery impacts customer satisfaction.

Author Contributions: Conceptualization, W.L. and J.P.; methodology, W.L. and J.P.; software, W.L. and J.P.; validation, W.L. and J.P.; formal analysis, W.L. and J.P.; investigation, W.L. and J.P.; resources, W.L.; data curation, W.L. and J.P.; writing—original draft preparation, J.P.; writing—review and editing, W.L. and J.P.; visualization, W.L. and J.P.; supervision, W.L.; project administration, W.L.; funding acquisition, W.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (No. RS-2025-00556289).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Publicly available datasets were analyzed in this study. This data can be found here: http://www.airlinequality.com/ (accessed on 17 July 2023). The processed data presented in this study are available on request from the corresponding author due to third-party restrictions.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Kim, H.; So, K.K.F. The evolution of service failure and recovery research in hospitality and tourism: An integrative review and future research directions. *Int. J. Hosp. Manag.* **2023**, *111*, 103457. [CrossRef]
2. McCollough, M.A.; Berry, L.L.; Yadav, M.S. An empirical investigation of customer satisfaction after service failure and recovery. *J. Serv. Res.* **2000**, *3*, 121–137. [CrossRef]
3. Abeler, J.; Calaki, J.; Andree, K.; Basek, C. The power of apology. *Econ. Lett.* **2010**, *107*, 233–235. [CrossRef]
4. Ozuem, W.; Ranfagni, S.; Willis, M.; Rovai, S.; Howell, K. Exploring customers' responses to online service failure and recovery strategies during COVID-19 pandemic: An actor-network theory perspective. *Psychol. Mark.* **2021**, *38*, 1440–1459. [CrossRef]
5. Radu, A.G.; Arli, D.; Surachartkumtonkun, J.; Weaven, S.; Wright, O. Empathy and apology: The effectiveness of recovery strategies. *Mark. Intell. Plan.* **2019**, *37*, 358–371. [CrossRef]
6. Wirtz, J.; Mattila, A.S. Consumer responses to compensation, speed of recovery and apology after a service failure. *Int. J. Serv. Ind. Manag.* **2004**, *15*, 150–166. [CrossRef]
7. Park, J.; Yoo, J.W.; Cho, Y.; Park, H. Examining the impact of service robot communication styles on customer intimacy following service failure. *J. Retail. Consum. Serv.* **2023**, *75*, 103511. [CrossRef]
8. Toney, D.E.; Hayes, L.J. A behavioral analysis of apologies, forgiveness, and interpersonal conflict. *Behav. Soc. Issues* **2017**, *26*, 128–155. [CrossRef]
9. Tsarenko, Y.; Tojib, D. Consumers' forgiveness after brand transgression: The effect of the firm's corporate social responsibility and response. *J. Mark. Manag.* **2015**, *31*, 1851–1877. [CrossRef]
10. Kim, T.T.; Kim, W.G.; Kim, H.-B. The effects of perceived justice on recovery satisfaction, trust, word-of-mouth, and revisit intention in upscale hotels. *Tour. Manag.* **2009**, *30*, 51–62. [CrossRef]
11. Grégoire, Y.; Fisher, R.J. The effects of relationship quality on customer retaliation. *Mark. Lett.* **2006**, *17*, 31–46. [CrossRef]
12. Grégoire, Y.; Fisher, R.J. Customer betrayal and retaliation: When your best customers become your worst enemies. *J. Acad. Mark. Sci.* **2008**, *36*, 247–261. [CrossRef]
13. Lee, J.-S.; Kim, J.; Hwang, J.; Cui, Y. Does love become hate or forgiveness after a double deviation? The case of hotel loyalty program members. *Tour. Manag.* **2021**, *84*, 104279. [CrossRef]
14. Lteif, L.; Rubin, D.; Ball, J.; Lamberton, C. There's not much to tell: The impact of emotional resilience on negative word-of-mouth following service failure. *Psychol. Mark.* **2023**, *40*, 1808–1820. [CrossRef]
15. Obeidat, Z.M.I.; Xiao, S.H.; Iyer, G.R.; Nicholson, M. Consumer Revenge Using the Internet and Social Media: An Examination of the Role of Service Failure Types and Cognitive Appraisal Processes. *Psychol. Mark.* **2017**, *34*, 496–515. [CrossRef]
16. Zhang, Y.; Zhang, J.; Sakulsinlapakorn, K. Love becomes hate? Or love is blind? Moderating effects of brand love upon consumers' retaliation towards brand failure. *J. Prod. Brand Manag.* **2020**, *30*, 415–432. [CrossRef]
17. Lin, C.-Y.; Chou, E.-Y. Investigating the role of customer forgiveness following a double deviation. *J. Serv. Mark.* **2022**, *36*, 1042–1057. [CrossRef]
18. Martinez, L.M.; Pacheco, N.; Ramos, F.R.; Bicho, M. Would you try it again? Dual effects of customer mindfulness on service recovery. *J. Retail. Consum. Serv.* **2023**, *74*, 103438. [CrossRef]
19. Bitner, M.J.; Booms, B.H.; Tetreault, M.S. The service encounter: Diagnosing favorable and unfavorable incidents. *J. Mark.* **1990**, *54*, 71–84. [CrossRef]

20. Shen, L. United Airlines Stock Drops $1.4 Billion After Passenger Removal Controversy. Fortune. 2017. Available online: https://fortune.com/2017/04/11/united-airlines-stock-drop/ (accessed on 12 January 2026).
21. McCann, E. United's Apologies: A Timeline. The New York Times. 2017. Available online: https://www.nytimes.com/2017/04/14/business/united-airlines-passenger-doctor.html (accessed on 12 January 2026).
22. Althubaiti, A. Information bias in health research: Definition, pitfalls, and adjustment methods. *J. Multidiscip. Healthc.* **2016**, *9*, 211–217. [CrossRef] [PubMed]
23. Kingstone, A.; Smilek, D.; Eastwood, J.D. Cognitive ethology: A new approach for studying human cognition. *Br. J. Psychol.* **2008**, *99*, 317–340. [CrossRef]
24. Berezina, K.; Bilgihan, A.; Cobanoglu, C.; Okumus, F. Understanding satisfied and dissatisfied hotel customers: Text mining of online hotel reviews. *J. Hosp. Mark. Manag.* **2016**, *25*, 1–24. [CrossRef]
25. Grégoire, Y.; Mattila, A.S. Service Failure and Recovery at the Crossroads: Recommendations to Revitalize the Field and its Influence. *J. Serv. Res.* **2020**, *24*, 323–328. [CrossRef]
26. Bhattacherjee, A. Understanding Information Systems Continuance: An Expectation-Confirmation Model. *MIS Q.* **2001**, *25*, 351–370. [CrossRef]
27. Bowen, J.T.; Chen, S.L. The relationship between customer loyalty and customer satisfaction. *Int. J. Contemp. Hosp. Manag.* **2001**, *13*, 213–217. [CrossRef]
28. Um, S.; Chon, K.; Ro, Y. Antecedents of revisit intention. *Ann. Tour. Res.* **2006**, *33*, 1141–1158. [CrossRef]
29. Anderson, E.W. Customer Satisfaction and Word of Mouth. *J. Serv. Res.* **1998**, *1*, 5–17. [CrossRef]
30. Zeelenberg, M.; Pieters, R. Beyond valence in customer dissatisfaction: A review and new findings on behavioral responses to regret and disappointment in failed services. *J. Bus. Res.* **2004**, *57*, 445–455. [CrossRef]
31. Azemi, Y.; Ozuem, W.; Howell, K.E. The effects of online negative word-of-mouth on dissatisfied customers: A frustration-aggression perspective. *Psychol. Mark.* **2020**, *37*, 564–577. [CrossRef]
32. Chang, J.R.; Chen, M.Y.; Chen, L.S.; Tseng, S.C. Why Customers Don't Revisit in Tourism and Hospitality Industry? *IEEE Access* **2019**, *7*, 146588–146606. [CrossRef]
33. Zhou, G.; Liao, C. Dynamic Measurement and Evaluation of Hotel Customer Satisfaction Through Sentiment Analysis on Online Reviews. *J. Organ. End User Comput.* **2021**, *33*, 1–27. [CrossRef]
34. Park, J. Combined Text-Mining/DEA method for measuring level of customer satisfaction from online reviews. *Expert Syst. Appl.* **2023**, *232*, 120767. [CrossRef]
35. Sann, R.; Lai, P.-C. Understanding homophily of service failure within the hotel guest cycle: Applying NLP-aspect-based sentiment analysis to the hospitality industry. *Int. J. Hosp. Manag.* **2020**, *91*, 102678. [CrossRef]
36. Borghi, M.; Mariani, M.M.; Vega, R.P.; Wirtz, J. The impact of service robots on customer satisfaction online ratings: The moderating effects of rapport and contextual review factors. *Psychol. Mark.* **2023**, *40*, 2355–2369. [CrossRef]
37. Bridges, J.; Vásquez, C. If nearly all Airbnb reviews are positive, does that make them meaningless? *Curr. Issues Tour.* **2018**, *21*, 2057–2075. [CrossRef]
38. Qiu, L.; Pang, J.; Lim, K.H. Effects of conflicting aggregated rating on eWOM review credibility and diagnosticity: The moderating role of review valence. *Decis. Support Syst.* **2012**, *54*, 631–643. [CrossRef]
39. Gregoriades, A.; Pampaka, M.; Herodotou, H.; Christodoulou, E. Explaining tourist revisit intention using natural language processing and classification techniques. *J. Big Data* **2023**, *10*, 60. [CrossRef]
40. Lin, C.-H.; Nuha, U. Sentiment analysis of Indonesian datasets based on a hybrid deep-learning strategy. *J. Big Data* **2023**, *10*, 88. [CrossRef] [PubMed]
41. Weun, S.; Beatty, S.E.; Jones, M.A. The impact of service failure severity on service recovery evaluations and post-recovery relationships. *J. Serv. Mark.* **2004**, *18*, 133–146. [CrossRef]
42. Zeithaml, V.A.; Berry, L.L.; Parasuraman, A. The nature and determinants of customer expectations of service. *J. Acad. Mark. Sci.* **1993**, *21*, 1–12. [CrossRef]
43. Lee, E.J.; Park, J. Service failures in online double deviation scenarios: Justice theory approach. *Manag. Serv. Qual. Int. J.* **2010**, *20*, 46–69. [CrossRef]
44. Smith, A.K.; Bolton, R.N.; Wagner, J. A model of customer satisfaction with service encounters involving failure and recovery. *J. Mark. Res.* **1999**, *36*, 356–372. [CrossRef]
45. Vázquez-Casielles, R.; Suárez Álvarez, L.; Díaz Martín, A.M. Perceived justice of service recovery strategies: Impact on customer satisfaction and quality relationship. *Psychol. Mark.* **2010**, *27*, 487–509. [CrossRef]
46. Le, A.N.H.; Ho, H.X. The behavioral consequences of regret, anger, and frustration in service settings. *J. Glob. Mark.* **2020**, *33*, 84–102. [CrossRef]
47. Cugueró-Escofet, N.; Fortin, M.; Canela, M.-A. Righting the wrong for third parties: How monetary compensation, procedure changes and apologies can restore justice for observers of injustice. *J. Bus. Ethics* **2014**, *122*, 253–268. [CrossRef]

48. Fu, H.; Wu, D.C.; Huang, S.S.; Song, H.; Gong, J. Monetary or nonmonetary compensation for service failure? A study of customer preferences under various loci of causality. *Int. J. Hosp. Manag.* **2015**, *46*, 55–64. [CrossRef]
49. Grewal, D.; Roggeveen, A.L.; Tsiros, M. The effect of compensation on repurchase intentions in service recovery. *J. Retail.* **2008**, *84*, 424–434. [CrossRef]
50. Roschk, H.; Kaiser, S. The nature of an apology: An experimental study on how to apologize after a service failure. *Mark. Lett.* **2013**, *24*, 293–309. [CrossRef]
51. Roschk, H.; Gelbrich, K. Identifying appropriate compensation types for service failures: A meta-analytic and experimental analysis. *J. Serv. Res.* **2014**, *17*, 195–211. [CrossRef]
52. Shi, Y.; Zhang, R.; Ma, C.; Wang, L. Robot service failure: The double-edged sword effect of emotional labor in service recovery. *J. Serv. Theory Pract.* **2022**, *33*, 72–88. [CrossRef]
53. Valentini, S.; Orsingher, C.; Polyakova, A. Customers' emotions in service failure and recovery: A meta-analysis. *Mark. Lett.* **2020**, *31*, 199–216. [CrossRef]
54. Adams, J.S. Inequity in social exchange. In *Advances in Experimental Social Psychology*; Academic Press: Cambridge, MA, USA, 1965; Volume 2, pp. 267–299. [CrossRef]
55. Nazifi, A.; Gelbrich, K.; Grégoire, Y.; Koch, S.; El-Manstrly, D.; Wirtz, J. Proactive handling of flight overbooking: How to reduce negative eWOM and the costs of bumping customers. *J. Serv. Res.* **2021**, *24*, 206–225. [CrossRef]
56. Wang, C.; Mattila, A.S. A cross-cultural comparison of perceived informational fairness with service failure explanations. *J. Serv. Mark.* **2011**, *25*, 429–439. [CrossRef]
57. Sparks, B.; Fredline, L. Providing an explanation for service failure: Context, content, and customer responses. *J. Hosp. Tour. Res.* **2007**, *31*, 241–260. [CrossRef]
58. Uzir, M.U.H.; Al Halbusi, H.; Thurasamy, R.; Thiam Hock, R.L.; Aljaberi, M.A.; Hasan, N.; Hamid, M. The effects of service quality, perceived value and trust in home delivery service personnel on customer satisfaction: Evidence from a developing country. *J. Retail. Consum. Serv.* **2021**, *63*, 102721. [CrossRef]
59. Holloway, B.B.; Beatty, S.E. Service failure in online retailing: A recovery opportunity. *J. Serv. Res.* **2003**, *6*, 92–105. [CrossRef]
60. Jeon, S.; Kim, J.S. Effects of service failure on consumer responses across failure types: A moderating role of intimacy. *Australas. Mark. J.* **2016**, *24*, 46–53. [CrossRef]
61. Zou, S.; Migacz, S.J. Why service recovery fails? Examining the roles of restaurant type and failure severity in double deviation with justice theory. *Cornell Hosp. Q.* **2022**, *63*, 169–181. [CrossRef]
62. Bies, R.J.; Shapiro, D.L. Interactional fairness judgments: The influence of causal accounts. *Soc. Justice Res.* **1987**, *1*, 199–218. [CrossRef]
63. Wang, X.; Zheng, J.; Tang, L.; Luo, Y. Recommend or not? The influence of emotions on passengers' intention of airline recommendation during COVID-19. *Tour. Manag.* **2023**, *95*, 104675. [CrossRef]
64. Anderson, S.W.; Baggett, L.S.; Widener, S.K. The Impact of Service Operations Failures on Customer Satisfaction: Evidence on How Failures and Their Source Affect What Matters to Customers. *Manuf. Serv. Oper. Manag.* **2008**, *11*, 52–69. [CrossRef]
65. Mikolov, T.; Chen, K.; Corrado, G.; Dean, J. Efficient estimation of word representations in vector space. *arXiv* **2013**, arXiv:1301.3781. [CrossRef]
66. Ahuja, S.; Dubey, G. Clustering and sentiment analysis on Twitter data. In Proceedings of the IEEE International Conference on Telecommunication and Networks (TEL-NET), Noida, India, 10–11 August 2017; pp. 1–5. [CrossRef]
67. Dao, H.M.; Theotokis, A. Self-Service Technology Recovery: The Effect of Recovery Initiation and Locus of Responsibility. *J. Interact. Mark.* **2021**, *54*, 25–39. [CrossRef]
68. Weiner, B. An attributional theory of achievement motivation and emotion. *Psychol. Rev.* **1985**, *92*, 548. [CrossRef]
69. Folkes, V.S. Consumer reactions to product failure: An attributional approach. *J. Consum. Res.* **1984**, *10*, 398–409. [CrossRef]
70. Hu, Y.; Min, H.; Su, N. How sincere is an apology? Recovery satisfaction in a robot service failure context. *J. Hosp. Tour. Res.* **2021**, *45*, 1022–1043. [CrossRef]
71. Burgoon, J.K. Interpersonal expectations, expectancy violations, and emotional communication. *J. Lang. Soc. Psychol.* **1993**, *12*, 30–48. [CrossRef]
72. Kwon, S.; Jang, S.S. Effects of compensation for service recovery: From the equity theory perspective. *Int. J. Hosp. Manag.* **2012**, *31*, 1235–1243. [CrossRef]
73. Suresh, P.; Chawla, V. The Burden of Double Deviation in Services: A Systematic Review and Research Agenda. *Int. J. Consum. Stud.* **2022**, *46*, 1919–1941. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.